

Hysterosalpingography in the evaluation of infertility.

PMID: 6226176 | Indexed in PubMed

Hysterosalpingography (HSG) was performed in 220 women for evaluation of infertility. Laparoscopy or laparotomy for final assessment of tubal patency were carried out in 110. HSG revealed a normal tubal patency in 66.8 per cent, whereas abnormal patency was found in 24.1 per cent. At operation, 85.7 per cent of the HSG diagnoses were confirmed. A clear correlation existed between previous lesions in the genital tract and pathologic findings at HSG and operation. The pregnancy rate following HSG was 15.4 per cent. It is concluded that HSG is of great value in the initial evaluation of infertility. The findings facilitate the decision regarding the final operative procedure: laparoscopy or laparotomy.

Procedural and clinical outcomes of Adiana® hysteroscopic tubal occlusion in the Netherlands.

PMID: 39718327 | Indexed in PubMed

The Adiana® Permanent Contraception System was a hysteroscopic tubal occlusion device but was withdrawn from the market in 2012. To evaluate the safety, feasibility and efficacy of the Adiana hysteroscopic tubal occlusion. A prospective observational multicentre cohort study of 300 women undergoing hysteroscopic sterilisation using the Adiana® was conducted in the Netherlands between 2009 and 2012. All procedures were performed using the same study protocol. Three months after bilateral placement a hysterosalpingography (HSG) was performed to confirm tubal occlusion. In 2018-2020 follow-up questionnaires were sent to all women. The primary outcome was the success rate of the Adiana tubal occlusion technique. Successful tubal occlusion was defined as an uneventful procedure with occluded fallopian tubes according to the HSG after 3 months. Secondary outcomes were the success rate of the device placement, the number of complications during placement and the pregnancy rate. Bilateral placement of Adiana devices was achieved in 93.5% of cases. Bilateral confirmed occlusion by HSG was accomplished in 87.9% of cases with successful Adiana placement. This was 77.1% in the intention-to-treat group. Complications and side effects were reported in 4.4% of women. The pregnancy rate was 3.6% in women with proven bilateral tubal occlusion. Hysteroscopic tubal occlusion using the Adiana technology is associated with a pregnancy rate of 3.6%. Although this technology was removed from the commercial market, this evaluation of the Adiana technology could provide useful information for the development of potential new, more effective hysteroscopic tubal occlusion devices. Hysteroscopic tubal occlusion techniques are no longer available on the market. This evaluation of Adiana could provide useful information for the development of potential new hysteroscopic tubal occlusion devices.

Evaluating Lipiodol Efficacy in Proximal Tubal Occlusion Treatment: Single-Center Experience and Literature Review.

PMID: 39654272 | Indexed in PubMed

BACKGROUND Tubal infertility caused by proximal occlusion of fallopian tubes is a common cause of female infertility. Lipiodol (ethiodized oil) is an oil-based iodinated contrast medium, previously used in hysterosalpingography, and is now used as a therapeutic agent. This single-center study aimed to

evaluate the experience of outcomes from the use of Lipiodol in 96 women with infertility due to fallopian tube occlusion. **MATERIAL AND METHODS** In total, 96 women with infertility caused by tubal obstruction, confirmed in hysterosalpingography and/or laparoscopy, who underwent tubal recanalization followed by Lipiodol injection were enrolled. Information regarding the reproductive outcome at a minimum of 6 months following the treatment was collected by a telephone survey or direct contact. Pregnancy rate was calculated. **RESULTS** Average age on admission was 33 ± 4.1 years. In most cases (79%, 76/96), infertility was primary. Average duration of infertility was 2.8 years. Comorbidities included endocrinopathies and endometriosis. Complete patency was achieved in 96% of cases (92/96 patients). In 4 cases (4%), distal occlusion of 1 ovary was observed, resulting in partial procedural success. Tubal catheterization was needed in 29 patients (30%). No procedural complications were noted. Thirty-four patients conceived, resulting in a 35.8% pregnancy rate. Miscarriage was noted in 5 patients, and tubal ectopic pregnancy in 1 patient. No long-term complications were reported. **CONCLUSIONS** Our results show that use of oil-based contrast for tubal recanalization is safe and effective treatment for proximal tubal occlusion, with a relatively high pregnancy rate (36%). Further studies are required to fully discover its potential benefits.

vNOTES chromopertubation: a new method for assessing tubal patency and peritubal anatomy.

PMID: 39616607 | Indexed in PubMed

We conducted a retrospective cohort study of women with suspected tubal factor infertility who underwent vaginal natural orifice endoluminal surgery (vNOTES) chromopertubation and adnexal anatomy assessment at a large referral hospital. Twelve women aged between 29 and 38 years with suspected tubal factor infertility who underwent vNOTES chromopertubation were retrospectively examined. Demographic data, as well as clinical and surgical characteristics, were reviewed and the surgical technique was defined. The mean BMI was 34.2 ± 7.12 kg/m² (range 24.7-42.6). No conversion to laparotomy or laparoscopy was required. The median operative time was 39 (26.5-42) minutes. The median VAS scores at six and 12 h postoperatively were 3.5 (3-4) and 2 (1-2), respectively. There were no complications and all patients were discharged at the 12th postoperative hour. Considering the increased risk of surgical morbidity and mortality associated with increasing BMI and the difficulty in gaining access to the abdominal cavity not only in obese women but also in cases with previous extensive midline incisions, where dense intra-abdominal adhesions are likely, or in cases where esthetic considerations are important, vNOTES appears to be a useful alternative to assess tubal patency and adnexal anatomy and even to perform concomitant adnexal surgery.

Tubal flushing with oil-based contrast during hysterosalpingography versus tubal flushing by hysterosalpingo-foam sonography in infertile women undergoing fertility work-up: study protocol of a randomised controlled trial (FOil study).

PMID: 39581724 | Indexed in PubMed

Hysterosalpingography (HSG) and hysterosalpingo-foam sonography (HyFoSy) are commonly used tubal patency tests during the fertility work-up. Besides its diagnostic purpose, HSG with oil-based contrast can also be applied for its fertility-enhancing effect, by tubal flushing. HyFoSy is considered as less painful

compared with HSG, it lacks exposure to iodinated contrast medium and ionising radiation. The fertility-enhancing effects of HyFoSy are less studied and randomised controlled trials comparing pregnancy rates after HSG and HyFoSy are lacking. This study (FOil study) is initiated to compare the effectiveness of tubal flushing during HSG with oil-based contrast and HyFoSy. The FOil study is a nationwide, multicentre, open label, randomised controlled trial with a superiority design. Infertile women with an indication for tubal patency testing during their fertility work-up will be randomly assigned to HSG with oil-based contrast medium or HyFoSy. The primary outcome is conception within 6 months after randomisation leading to live birth. To demonstrate or refute an 8% difference in conception leading to live birth in favour of HSG with oil-based contrast, 1102 women will be included in the trial. A cost-effectiveness analysis from a societal perspective will be performed alongside the trial. The trial is approved by the Medical Ethics Review Committee of the Amsterdam University Medical Centers (Ref. No. 2022.0884, date: 17 March 2023) and by the boards of the participating hospitals. The findings will be disseminated in peer-reviewed journals and participants will be informed through the patient organisation. NCT05882188.

Imaging techniques for assessment of tubal status.

PMID: 24829524 | Indexed in PubMed

Fallopian tubes make a vital portal for transfer of gametes and embryo. Tubal factor is responsible for infertility in 25-35% of cases. Hysterosalpingography (HSG) has been used to evaluate the uterine cavity and the tubal status since decades. It uses iodinated contrast and X-rays and is painful and inconvenient for patient. Laparoscopy is considered to be the gold standard for tubal evaluation, but is an operative procedure and needs anesthesia. Though ultrasound is a modality of choice for assessment of uterus and ovaries, it does not allow assessment of the fallopian tube unless there is any fluid surrounding it or inside the lumen. This fluid interface can be created artificially by introducing saline in the uterine cavity and fallopian tubes and scanning simultaneously. The procedure is named Saline infusion HSG. Saline infusion sonohysterosalpingography (SIS) can be done with B mode US and Doppler. SIS can demonstrate a patent tube but if blocked, the site of block cannot be demonstrated. Ultrasound contrast agents can be used for tubal assessment using contrast mode on the scanners. This procedure is known as hystero-contrast sonography (HyCoSy). This actually shows the passage of hyperechoic contrast agent through tubal lumen and delineates it and locates the site of block. Using the volume ultrasound may even make the demonstration of tubal status and fimbriae better. Results of HyCoSy have been found to correlate well with laparoscopic findings, which are a gold standard. It is recommended by National Institute for Health and Clinical Excellence as a primary investigation for tubal assessment in patients without any positive history of tubal damage and also can replace a second look laparoscopy.

Appearance of uterine scar due to previous cesarean section on hysterosalpingography: various shapes, locations and sizes.

PMID: 24046789 | Indexed in PubMed

Hysterosalpingography (HSG) is the radiographic evaluation of the uterus and fallopian tubes that is used predominantly in the assessment of infertility and evaluation of abnormalities of the uterus and fallopian tubes. Some of the abnormalities that can be detected by HSG include congenital anomalies, polyps,

leiomyomas, synechiae and adenomyosis. HSG is also used to evaluate any scarring on the uterus and fallopian tubes. Cesarean section is the most commonly performed surgical procedure involving the uterus in fertile women. Cesarean section involves an incision made in the lower uterine segment or isthmus. Various changes in the site of the cesarean incision may be seen due to wall weakness and fibrosis. The scar may have various shapes; unilateral or bilateral, single or multiple, wedge-shaped or linear. Awareness of the appearance and locations of uterine defects due to previous cesarean section is necessary in order to differentiate them from normal variations and other pathologies mimicking it. In this study, we demonstrate the appearance of anatomic defects of the uterine cavity on HSG after cesarian section. We define different shapes such as thin linear defect, focal saccular outpouching, unilateral or bilateral diverticula (dog-ear like) and fistula and different locations such as the uterine body, lower uterine segment, uterine isthmus and the upper endocervical canal.

Hysterosalpingography.

PMID: 1853510 | Indexed in PubMed

During the past decade, there has been a dramatic increase in the number of women seeking infertility evaluation. Hysterosalpingography (HSG) is an invaluable procedure for evaluating internal architecture of the female reproductive tract. Utilizing HSG, it may be possible to minimize the use of invasive procedures, such as hysteroscopy and laparoscopy, in defining such problems as peritubal adhesions, leiomyomas, and congenital anomalies. This review re-emphasizes the wide range of available information and advantages of HSG which can be extremely useful in the diagnosis and management of infertile patients.

Imaging in Infertile Female Patients Who Underwent Hysterosalpingography Investigation at Dhulikhel Hospital.

PMID: 28814690 | Indexed in PubMed

Background Fallopian tube and uterine abnormalities are the most common cause of female infertility, accounting for 30% of cases. Hysterosalpingography is the safe, minimal invasive radiographic technique for evaluation of uterine cavity and fallopian tubes defects. The purpose of the study was to assess the various uterine and tubal abnormalities leading to infertility Objective To assess the abnormalities in uterus and fallopian tubes detected on Hysterosalpingography (as causative factors of infertility in a tertiary care centre for proper decision making in treatment. Method This is the prospective cross sectional study done in department of radiology Dhulikhel hospital, Kathmandu University Hospital between January 2015 to January 2016 with complain of infertility. Hysterosalpingography was performed using ionic contrast medium (10-20 ml of 76% urografin) under digital fluoroscopy. Radiographic films were obtained and analyzed. Demographic data and radiological findings were reviewed and the obtained data analyzed with SPSS version 16. Result Out of the total 100 patient, 75 had primary infertility while 25 had secondary infertility. The age ranged from 19 to 43 years with mean age of 27.75 years. Normal Hysterosalpingography findings were seen in 37% cases. Tubal abnormalities were seen in 34%, uterine abnormalities in 23% and both tubal and uterine abnormalities in 6% patients. The most common tubal abnormality detected on Hysterosalpingography was tubal block 75%. Out of the 30 patients who had tubal block, unilateral block was noted in 76.7% patients whereas bilateral block was

noted in 23.3%. Hydrosalpinx was seen in 25% cases with tubal abnormality. Among the 29 cases with various uterine abnormalities, bicornuate uterus 9% was the most common abnormality Conclusion Hysterosalpingography is an easily available radiographic procedure that can demonstrate a wide variety of uterine and tubal abnormalities for the initial assessment of infertility which can avoid unnecessary and sometimes more aggressive procedures.

[Comparison of current methods of tubal patency assessment].

PMID: 28245679 | Indexed in PubMed

Most common organic cause of infertility is the blockage of the Fallopian tubes. Several methods were introduced to evaluate tubal patency. Hysterosalpingography is a conventional radiology procedure using contrast medium, which gives an accurate image of the uterine cavity and the Fallopian tubes, but radiation exposure is necessary. Hystero-contrast-sonography similarly examines the uterine cavity and tubal patency by ultrasonography, and it enables to detect pelvic pathology, too. Transvaginal hydrolaparoscopy is a minimal invasive direct method using endoscope introduced into the abdominal cavity through the posterior vaginal fornix, both ovaries and tubal patency can be observed. Laparoscopy is the "gold standard" procedure in the tubal testing, however it is a more invasive procedure. A cost-effective testing method is the selective tubal pertubation performed via office hysteroscopy. Recent outpatient methods to detect tubal patency have high negative predictive values and recommended to be the first choice in infertility work-up. Orv. Hetil., 2017, 158(9), 324-330.
